Certificate-driven polytope-hull contraction for interval branch and bound

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Abstract

Interval branch-and-bound solvers compute rigorous enclosures of the solutions of nonlinear systems and of global optima. Their polytope-hull contractor bounds each of the n variables of a node over a linear relaxation by solving $2n$ small linear programs (LPs) with a floating-point LP solver, whose dual solutions are certified afterwards by the Neumaier–Shcherbina bound. That bound is valid for any multipliers, so any LP solver is safe to use; yet a general-purpose solver treats each of the $2n$ LPs as a new problem. We present a self-certifying simplex (SC) dedicated to the pass: it solves the $2n$ LPs in sequence on one basis, updating only what changes from one LP to the next. The certificate makes this simple: any basis yields a valid bound and can be made dual feasible, so SC never verifies, refactorizes before an optimum or runs a phase 1. A ladder of configurations in Ibex, from SoPlex as Ibex uses it to SC, attributes the gain to each step. On recorded passes, from the same boxes and relaxations, a pass of SC on relaxations of up to 30 rows costs 0.39–0.42 times that of SoPlex as Ibex uses it, and 20–25% less than the simplex of SoPlex driven directly, with the same contraction. Inside the search, the optimizer takes about 20% less CPU time than with SoPlex as Ibex uses it. On larger relaxations SC is slower, and a hybrid that chooses the engine by the size of the relaxation keeps the best of both.

1 Introduction

Interval branch-and-bound solvers compute, with mathematical certainty, an enclosure of all solutions of a system of nonlinear equations and inequalities, or the global optimum of a constrained problem [20]. Their efficiency depends on *contractors*, operators that shrink the current box without losing solutions [7]. Among them, contractors based on linear relaxations are among the most effective: the constraints are linearized over the box, by convex interval Taylor forms [3] or affine arithmetic [23, 2], and each variable is minimized and maximized over the resulting polytope. In Ibex this is the polytope-hull contractor, the linear part of X-Newton, which we introduced in [3]; we later combined the affine and Taylor-based relaxations [2].

Each call solves up to $2n$ linear programs, and the rigor of the result cannot come from the LP solver, which works in floating point. Current implementations therefore treat the LP solver as a black box and certify its output a posteriori: the dual solution is turned into a safe bound by

the Neumaier–Shcherbina procedure [21], and an infeasibility claim is accepted only if the returned Farkas ray passes the same verification. This design has a cost that is easy to miss: the $2n$ LPs of a node are small and closely related, typically two or three pivots apart, and each is a separate call to a general-purpose solver, which prepares the LP as a new problem, rebuilding its vectors, and refactorizes the basis and checks its solution before declaring an optimum; on these LPs that work is a large part of the cost of a call (Section 5).

The certificate already makes any LP solver safe to use: with the relaxation written in a homogeneous form, with one interval-bounded slack per row, *any* multiplier vector yields a valid bound on the box and on every sub-box, whatever solver produced it. A black box cannot take advantage of this, because it treats each call as a new problem. We design the simplex for the pass instead: the relaxation is loaded once, and the $2n$ LPs are solved in sequence on one basis, keeping the factorization and updating only what the new objective and the contracted box change. The certificate is what makes such a simplex simple. Its multipliers never need to be correct, so it performs no verification of its own and does not refactorize before declaring an optimum; and every basis can be made dual feasible by relocating its nonbasic variables, so no LP of the pass, and no LP after the box contracts, needs a phase 1 (the search for a feasible starting basis). We call this simplex *self-certifying* (SC). Against the simplex of SoPlex driven directly, which also keeps its basis from one LP to the next, we find that what SC saves is mostly the preparation of each LP as a new problem: switching off the refactorization and the checks of SoPlex recovers little. How much it is worth depends on the size of the relaxation and on what else the contractor does, and measuring that is part of the contribution.

Positioning. The certificate itself is not new: it is the safe bound of Neumaier and Shcherbina [21], already applied to filtering with linear relaxations [9, 24]. In those approaches the multipliers are a by-product of an external solver and certification is a post-processing step. Resuming a dual simplex inside an interval method goes back to Yamamura and coworkers [27, 28], for an LP emptiness test; SC extends that warm start to the $2n$ bound problems, with rigorous certification. Section 2 gives the details.

Contributions. The paper makes three contributions.

- *A simplex dedicated to the pass.* SC solves the $2n$ problems of a node in sequence on one basis, without preparing each as a new problem, without a phase 1, and without the verification and the refactorization before an optimum of a general-purpose solver (Section 4). The properties it relies on, validity of the certificate for any multipliers, exactness at an optimum and dual feasibility for free, follow from the Neumaier–Shcherbina bound and LP duality; we state them in Section 3.1 because the design rests on them, not as new results.

- *An attribution of the gain.* A ladder of configurations inside the same contractor of Ibex, from SoPlex as Ibex uses it, through the merging of twin rows (a simple preprocessing that halves the relaxation of square systems for every engine), a tuned SoPlex and its simplex driven directly, to SC and a hybrid, measures what each step contributes, separately on optimization problems and systems of equations and on small and large relaxations: on a bank of recorded passes, where every rung starts from the same boxes and relaxations, and inside the search, with one instance per family (Section 5).

- *The boundary of the approach.* SC pays on small relaxations, where the fixed cost of a call dominates (per pass, up to about 30 rows); beyond, the pivots dominate, and SoPlex makes

them cheaper by switching to a row representation of the basis that SC does not have. We measure that explanation, and give a hybrid contractor that chooses the engine per node by the size of the relaxation (Sections 5 and 6).

2 Background and related work

Interval solvers and linear relaxations. Interval branch and bound alternates bisection with contraction. Ibex, on which we build, composes constraint propagation (HC4 [6]), shaving-based consistencies such as ACID [22], the interval Newton operator for square systems [19], and contractors based on linear relaxations [7], whose relaxation comes from convex interval Taylor forms [3], from affine arithmetic [23], or from both [2]. QUAD [17] and GloptLab [9] use linear relaxations in the same way.

Rigorous bounds for linear programs. A floating-point LP solver returns approximate solutions, and two lines of work make its results rigorous. The first certifies them a posteriori: any approximate dual vector yields a safe bound once the residual is evaluated in interval arithmetic, and an approximate Farkas ray proves infeasibility under the same verification [21, 14, 15]. The second solves the LP exactly, in rational arithmetic [1], or to arbitrary precision by iterative refinement [12], as in exact mixed-integer programming [8]. Both treat the LP solver as given; we keep the a posteriori certificate and change the solver. Domes and Neumaier [9] applied the a posteriori bound to filtering with linear relaxations; in previous work [24] we wrote the relaxation in the homogeneous form used here and showed that the dual solutions of the bound LPs give the optimal preconditioner for an interval Gauss–Seidel projection, and Gleixner et al. [11] reuse the equivalent inequalities in sub-boxes. In all of them certification is a post-processing step, whose cost leads to applying it to a few promising variables [9] or only periodically [24]. Minimizing and maximizing each variable over a relaxation is also the optimization-based bound tightening of global optimization solvers, where rigor is not required [25, 5, 11].

Warm starts in interval methods. Yamamura and coworkers [27, 28] observed that the optimal tableau of the parent remains dual feasible after a bisection, and resume an LP emptiness test by the dual simplex method. The heuristic of skipping the bounds already attained by a primal point, used by the polytope-hull contractor of Ibex, comes from Baharev, Achterberg and Rév [4].

Simplex techniques. SC uses standard ingredients, described by Koberstein [16] for the dual method: the bounded-variable dual simplex with Harris' ratio test [13], and the revised method with a sparse LU factorization updated by the method of Forrest and Tomlin [10], as in SoPlex [26]. What is specific to our setting is their use for $2n$ closely related LPs over the same constraints, with a certificate attached to every basis.

3 The contractor

This section describes the contractor independently of the engine that solves its LPs: the relaxation and its certificate (Section 3.1), the pass over the bounds (Section 3.2) and the merging of twin rows (Section 3.3). Section 4 then describes SC, and Section 5 compares it with external LP solvers inside this same contractor. Only SC is new; the rest is the common ground on which the engines are compared.

3.1 The homogeneous relaxation and its certificate

Let $\mathbf{x} \in \mathbb{IR}^n$ be the current box. A linearizer returns a matrix $A \in \mathbb{R}^{m \times n}$ of floating-point numbers and an interval vector $\mathbf{b} \in \mathbb{IR}^m$, possibly with infinite bounds, such that

$$x \in \mathbf{x} \text{ is a solution } \implies Ax \in \mathbf{b}. \tag{1}$$

We use the two linearizers of our previous work [2]: the *affine* relaxation [23], and the *combined* relaxation (`compo`), which adds to the affine rows those of X-Newton [3], one per constraint and corner of the box. We introduce one slack per row, $s = Ax$, and write the relaxation in the homogeneous form

$$\bar{A}z = 0, \qquad z = (x, s) \in \mathbf{z} := \mathbf{x} \times \mathbf{s}, \qquad \bar{A} := [\, A \;\; -I_m \,], \tag{2}$$

where $\mathbf{s}_i := \mathbf{b}_i \cap \square(A_{i:}\,\mathbf{x})$ and $\square(\cdot)$ denotes evaluation in interval arithmetic. We call $P(\mathbf{x}) := \{z \in \mathbf{z} : \bar{A}z = 0\}$ the *relaxation polytope*.

Lemma 1. *If $x^* \in \mathbf{x}$ is a solution, then $(x^*, Ax^*) \in P(\mathbf{x})$.*

Proof. $A_{i:}x^* \in \mathbf{b}_i$ by (1) and $A_{i:}x^* \in \square(A_{i:}\mathbf{x})$ by the inclusion property of interval arithmetic; the equation holds by construction of s. □

The certificate combines the m rows of $\bar{A}z = 0$ linearly, with one *multiplier* per row, into a single equation $\gamma^T z = 0$ that every point of the polytope satisfies. Any vector of multipliers $\lambda \in \mathbb{R}^m$ gives a valid equation; the best ones for bounding z_k are the dual solutions of the LP that minimizes or maximizes z_k over the polytope, and the simplex of Section 4 produces them.

Definition 2 (Certificate)**.** For a vector of multipliers $\lambda \in \mathbb{R}^m$, let $\gamma \in \mathbb{IR}^{n+m}$ be an interval vector that contains $\gamma := \bar{A}^T\lambda$, computed in outward-rounded interval arithmetic. For $k \in \{1, \ldots, n+m\}$ with $0 \notin \gamma_k$, the *certified enclosure* of z_k is

$$\mathbf{C}_k(\lambda, \mathbf{z}) := -\Big(\textstyle\sum_{j \neq k} \gamma_j\, \mathbf{z}_j\Big) \Big/ \gamma_k,$$

and $\mathbf{C}_k(\lambda, \mathbf{z}) := \mathbb{R}$ when $0 \in \gamma_k$; and the *residual* is $\mathbf{R}(\lambda, \mathbf{z}) := \sum_j \gamma_j\, \mathbf{z}_j$.

The idea is a single line. Every point z of the polytope satisfies $\bar{A}z = 0$, hence $\gamma^T z = \lambda^T \bar{A} z = 0$ for any λ, that is, $\gamma_k z_k = -\sum_{j \neq k} \gamma_j z_j$. Bounding the right-hand side over the box in interval arithmetic, and dividing by γ_k, encloses z_k; and if the residual $\mathbf{R}$ excludes zero, no point of the box can satisfy $\gamma^T z = 0$, so the polytope is empty. A contraction step therefore replaces $\mathbf{z}_k$ by $\mathbf{z}_k \cap \mathbf{C}_k(\lambda, \mathbf{z})$, and an emptiness test checks $0 \notin \mathbf{R}(\lambda, \mathbf{z})$. All statements below hold for the floating-point A returned by the linearizer; rounding in γ and $\mathbf{C}_k$ is covered by outward rounding.

Proposition 3 (Validity)**.** *Let $\lambda \in \mathbb{R}^m$ be arbitrary. For every $z \in P(\mathbf{x})$ and every k with $0 \notin \gamma_k$, $z_k \in \mathbf{C}_k(\lambda, \mathbf{z})$; if $0 \notin \mathbf{R}(\lambda, \mathbf{z})$, then $P(\mathbf{x}) = \emptyset$ and $\mathbf{x}$ contains no solution. Both claims hold on every sub-box $\mathbf{x}' \subseteq \mathbf{x}$ with the matrix $\bar{A}$ computed on $\mathbf{x}$.*

Proof. For $z \in P(\mathbf{x})$ and the exact $\gamma = \bar{A}^T\lambda \in \gamma$, $\gamma^T z = \lambda^T \bar{A} z = 0$; both claims follow from the inclusion property of interval arithmetic and Lemma 1. The linearization on $\mathbf{x}$ remains valid on $\mathbf{x}'$, so the polytope over $\mathbf{x}'$ lies in $P(\mathbf{x})$. □

No property of λ is used: it need not be optimal, feasible or accurate, and errors in it can only weaken the enclosure. Written back in the variables x, $\inf \mathbf{C}_k(\lambda, \mathbf{z})$ is the Neumaier–Shcherbina safe lower bound of $\min x_k$ subject to $Ax \in \mathbf{b}$, $x \in \mathbf{x}$, with approximate dual λ [21], and the row-wise solution of the preconditioned system of Domes and Neumaier [9, §5]; validity on sub-boxes is used by Gleixner et al. [11] and in our earlier work [24]. The homogeneous form, introduced in [24], adds one detail: each slack is bounded by $\square(A_{i:}\mathbf{x})$ even when a side of $\mathbf{b}_i$ is infinite, so a multiplier of the "wrong" sign on a one-sided row gives a weaker but finite bound, and literally any λ certifies. Two standard facts complete what the design uses. At an optimal basis of the bound problem $\mathrm{LP}_k^{\min}$ (see (3)) with z_k basic, $\inf \mathbf{C}_k(\lambda^*, \mathbf{z})$ equals the optimal value in exact arithmetic, by LP duality [24, Prop. 1]; in floating point the excess is at most $\sum_j \mathrm{wid}(\gamma_j)\,\mathrm{mag}(\mathbf{z}_j)$ plus the effect of the inaccuracy of λ, as for the Neumaier–Shcherbina bound. And every basis can be made dual feasible for free [16]:

Proposition 4 (Dual-feasible start)**.** *The slack basis $B = -I_m$ is dual feasible for the bound problem of every structural variable z_k, $k \le n$, with $\mathbf{z}_k$ bounded. If every $\mathbf{z}_j$ is bounded, any basis becomes dual feasible after placing each nonbasic z_j at its lower bound if its reduced cost $d_j > 0$ and at its upper bound if $d_j < 0$.*

Proof. With the slack basis the cost of every basic variable is zero, so $\lambda = 0$ and $d = e_k$, and z_k at its lower bound (at its upper bound for the maximization) has the right sign. In general d does not depend on where the nonbasic variables sit, and the relocation enforces the sign conditions whenever the required bound is finite. □

When the objective changes from one bound to the next, or the box contracts, relocation thus restores dual feasibility without a phase 1, which is what lets one basis serve the $2n$ problems; the case of a box change is the warm start of Yamamura and coworkers [27, 28]. Section 4 explains how the simplex produces λ.

Why not a rigorous simplex. Since any multipliers give a valid bound, rigor is needed only in the final evaluation of γ and $\mathbf{C}_k$, not in the pivoting. Pivoting in interval arithmetic pays the width growth of interval Gaussian elimination [19], and solving each LP exactly or to high precision [1, 12] buys an exactness that the bound does not need. On the root boxes of 153 instances, an early version of our contractor that pivoted in interval arithmetic took 2.8 times as long, with the same mean contraction (24.6% against 24.0%).

3.2 One pass over the bounds

At each node the relaxation is computed, written in the form (2), and loaded once. The contractor then makes one pass over the up to $2n$ bound problems

$$\mathrm{LP}_k^{\sigma}: \quad \sigma\, z_k \quad \text{s.t.} \quad \bar{A}z = 0,\ z \in \mathbf{z}, \qquad k \le n,\ \sigma \in \{\min, \max\}, \tag{3}$$

Algorithm 1 gives the pass. The set L holds the bounds still to solve (line 1). Each LP is warm-started from the final basis of the previous one, and the engine returns the multipliers of the basis where it stops (line 4). They certify the bound once, and the box is contracted before the next LP starts (line 6), so every later LP works on the smaller box. If the residual $\mathbf{R}(\lambda, \mathbf{z})$ excludes zero, the node is empty and the pass stops (line 5).

No other outcome stops the pass, because any multipliers certify (Proposition 3): an LP that stops at its iteration limit, or reports an infeasibility whose ray fails the residual test, still contracts

Algorithm 1 One pass over the bounds of a node. SOLVE is the LP engine, SC (Algorithm 2) or an external solver.

Require: relaxation $\bar{A}z = 0$ of the node (2), twin rows merged; box $\mathbf{z} = \mathbf{x} \times \mathbf{s}$
Ensure: the box $\mathbf{z}$ contracted, or $\emptyset$ if the node has no solution

1: $L \leftarrow \{(k, \sigma) : k \leq n,\ \sigma \in \{\min, \max\}\}$
2: **while** $L \neq \emptyset$ **do**
3: $\quad(k, \sigma) \leftarrow$ next bound of L; remove it from L ▷ skipping and order
4: $\quad\lambda \leftarrow \text{SOLVE}(\text{LP}_k^\sigma, \mathbf{z})$ ▷ warm start, at most 100 iterations
5: $\quad$**if** $0 \notin \mathbf{R}(\lambda, \mathbf{z})$ **then return** $\emptyset$ ▷ Proposition 3
6: $\quad\mathbf{z}_k \leftarrow \mathbf{z}_k \cap \mathbf{C}_k(\lambda, \mathbf{z})$ ▷ Neumaier–Shcherbina bound [21]
7: **return** $\mathbf{z}$

with the multipliers it returns, if less than an optimal basis would. The polytope-hull contractor of Ibex 2.8.9 instead abandons the pass in both cases; we corrected it, the correction has been submitted to the Ibex distribution, and every configuration of Section 5 uses it, including the one that stands for Ibex. Ibex limits SoPlex to 100 iterations per LP, and every engine of Section 5, SC included, uses that limit.

Line 3 chooses the next bound with two rules of the polytope-hull contractor of Ibex [4], which use the primal point of the last LP solved to optimality. A bound already attained by such a point is *skipped*: it is removed from L without solving its LP, which would return that value, and the pass ends when every remaining bound is skipped. Among the others, the next one is the bound nearest to the last primal point, whose optimum lies closest to the vertex where the previous LP ended, so that it needs fewer pivots.

3.3 Merging twin rows

Rows of the relaxation that are proportional, within a relative tolerance of 10^{-9}, are merged into one row whose slack interval is the intersection of their scaled intervals; the polytope does not change. The affine linearizer writes each equation as two one-sided rows with the same normal, so merging halves m on square systems, and `compo` writes some rows twice, with opposite signs, far apart in the matrix. Merging is done once per node, before any LP, and gives a smaller LP to SC and to an external solver alike.

4 A simplex for the $2n$ bound problems

For each structural variable x_k the pass of Section 3.2 needs the bound problems $\text{LP}_k^{\min}$ and $\text{LP}_k^{\max}$ (3). We write $[l_j, u_j] := \mathbf{z}_j$. A basis is a set of m columns of $\bar{A}$ forming a nonsingular matrix B; B_r is the variable basic in row r, x_B the vector of values of the basic variables, N the set of nonbasic variables and $\bar{A}_N$ their columns. With cost vector $c = \pm e_k$, the multipliers are $\lambda^T = c_B^T B^{-1}$ and the reduced costs $d = c - \bar{A}^T\lambda$. For the bounded-variable simplex, a basis with each nonbasic variable at one of its bounds is *dual feasible* if $d_j \geq 0$ for nonbasic variables at their lower bound and $d_j \leq 0$ for those at their upper bound, and *primal feasible* if the basic variables lie within their bounds. This section describes how the simplex solves the $2n$ problems of a node; the design choices are evaluated in Section 5.

A simplex for one pass. SC is not a general LP solver. It solves the $2n$ LPs of one pass over a relaxation loaded once, in sequence and on one basis: between two LPs it keeps its factorization and recomputes only what the new objective and the contracted box change. The certificate is what makes this simple. Since it is valid for any λ (Proposition 3), the simplex never has to return a correct solution, only a good basis. A general-purpose LP solver, before declaring an optimum, refactorizes the basis to remove the error accumulated by the updates of its factorization, recomputes its vectors and verifies the solution, and the contractor then verifies it once more through the certificate. SC uses the final basis as it is: an inaccurate λ yields a weaker bound, never an invalid one, so the error of the factorization is controlled only as far as pivoting requires. Likewise, since every basis can be made dual feasible (Proposition 4), no phase 1 is needed when the objective changes or the box contracts, and the $2n$ problems share one basis; with the primal warm start of Section 4.1 none is needed either, because the previous optimal basis stays primal feasible. Section 5.1 measures what this design is worth: against the simplex of SoPlex driven directly, the gain comes mostly from not preparing each LP as a new problem, more than from the omitted checks.

Algorithm 2 shows one call of SC, the SOLVE of Algorithm 1. Lines 2–8 prepare the LP from the state left by the previous one (Section 4.1). The loop of lines 9–31 combines both methods: it takes a dual step while some basic variable violates its bounds, a primal step while some reduced cost has the wrong sign, and stops at an optimal basis (line 25), at a Farkas ray (line 14) or at the iteration limit (line 31); Section 4.2 describes the steps. After a dual warm start the basis is dual feasible and only dual steps occur; after a primal one it is primal feasible and only primal steps occur. The two lines marked $\star$ are where the design described above departs from a general-purpose solver: line 4, the warm start, starts each LP from the final basis of the previous one, with its factors, and line 25 returns the basis as it is. The certificate is computed by the caller, in lines 5–6 of Algorithm 1, once per LP and over the rows with $\lambda_i \neq 0$; it costs the same as the Neumaier–Shcherbina post-processing of the dual of an external solver (1.3–3.1 μs per LP on three optimization problems).

4.1 Starting and warm-starting

The first LP of a pass starts from the slack basis $B = -I_m$ (line 2 of Algorithm 2), which is dual feasible for every LP_k^σ (Proposition 4), and is solved by the dual method, with no phase 1: the costs of the slacks are zero, so $\lambda = 0$ and $d = c$ (line 5); every nonbasic variable starts at its lower bound, and for a maximization line 7 moves z_k to its upper bound. Between two consecutive LPs the objective changes and the contracted bound changes the box. SC keeps the basis and its factors (line 4) and recomputes λ and d for the new objective (line 5); two warm starts are then possible. The *dual* one relocates every nonbasic variable to the bound indicated by the sign of its new reduced cost (line 7), which restores dual feasibility, and continues with the dual method; the *primal* one keeps the nonbasic variables where they are and reoptimizes with primal steps, as an LP solver does. Under the dual method the certificate is available at every iteration, and it never gets worse:

Proposition 5 (Monotone certified bound)**.** *Run the dual simplex method on* $\mathrm{LP}_k^{\min}$ *from a dual-feasible basis. At every iteration* t *in which* z_k *is basic,* $\inf \mathbf{C}_k(\lambda_t, \mathbf{z})$ *is a valid lower bound on* z_k *over* $P(\mathbf{x})$ *and, in exact arithmetic, it is nondecreasing in* t*.*

Proof. Validity is Proposition 3. With z_k basic, $d = e_k - \bar{A}^T\lambda_t$ has $d_k = 0$, so $\gamma_k = 1$, $\gamma_j = -d_j$ for $j \neq k$, and $\inf \mathbf{C}_k = \sum_{j\neq k} \min_{z_j \in \mathbf{z}_j} d_j z_j = \min_{z\in\mathbf{z}} d^T z$, the Lagrangian dual function of $\mathrm{LP}_k^{\min}$ at λ_t, which the iterates of the dual simplex method do not decrease. □

Algorithm 2 SOLVE in SC: one LP of the pass. Comments marked $\star$ point to the work a general-purpose solver does differently.

Require: relaxation $\bar{A}z = 0$, loaded once per node; bound (k, σ), with cost $c = e_k$ to minimize and $c = -e_k$ to maximize; current box $\mathbf{z}$; warm start, primal or dual (Section 4.1); except in the first LP of the node, the final basis B of the previous LP and its LU factors

Ensure: multipliers $\lambda \in \mathbb{R}^m$: those of an optimal basis, a Farkas ray, or those of the basis reached at the iteration limit

if first LP of the node **then**
  $B \leftarrow -I_m$, factorized; each nonbasic z_j at its lower bound ▷ Prop. 4
else
  keep B and its LU factors from the previous LP ▷ $\star$ a solver starts anew
$\lambda \leftarrow B^{-T} c_B$; $d \leftarrow c - \bar{A}^T \lambda$ ▷ multipliers and reduced costs for c
if first LP **or** dual warm start **then**
  move each nonbasic z_j whose d_j has the wrong sign to its other bound
$x_B \leftarrow -B^{-1} \bar{A}_N z_N$ ▷ values of the basic variables
for $t = 1, \ldots, 100$ **do**
  if some basic variable violates its bounds **then** ▷ dual step
    $r \leftarrow$ the basic variable with the largest violation; $\delta \leftarrow x_{B_r}$ minus the bound it violates
    $\rho_r \leftarrow B^{-T} e_r$; $\alpha_r^T \leftarrow \rho_r^T \bar{A}$
    $q \leftarrow$ ratio test on d_j / α_{rj} with Harris' rule
    if no q exists **then return** $\lambda \leftarrow \rho_r$ ▷ Farkas ray
    $\alpha_q \leftarrow B^{-1} \bar{A}_{:q}$; $\theta_p \leftarrow \delta / \alpha_{rq}$
  else if some nonbasic d_j has the wrong sign **then** ▷ primal step
    $q \leftarrow$ the nonbasic variable with the largest such $|d_j|$; $\alpha_q \leftarrow B^{-1} \bar{A}_{:q}$
    $\theta_p \leftarrow$ the largest step of z_q, of the sign of $-d_q$, that keeps x_B within bounds
    $r \leftarrow$ the basic variable that reaches its bound at θ_p ▷ stable entries of α_q only
    if z_q reaches its other bound first **then**
      $x_B \leftarrow x_B - (u_q - l_q) \operatorname{sign}(-d_q)\, \alpha_q$; move z_q to that bound
      go to the next iteration ▷ bound flip, no pivot
    $\rho_r \leftarrow B^{-T} e_r$; $\alpha_r^T \leftarrow \rho_r^T \bar{A}$
  else
    return λ ▷ optimal; $\star$ a solver refactorizes and verifies first
  $\theta_d \leftarrow d_q / \alpha_{rq}$
  $x_B \leftarrow x_B - \theta_p \alpha_q$; $\lambda \leftarrow \lambda + \theta_d \rho_r$; $d \leftarrow d - \theta_d \alpha_r$
  $B_r \leftarrow q$; $x_{B_r} \leftarrow z_q + \theta_p$ ▷ the leaving variable stays at its bound
  update the LU factors of B ▷ Forrest–Tomlin
  if 50 pivots since the last factorization **then** refactorize B
return λ ▷ iteration limit; λ still certifies

Both warm starts reach an optimal basis and give the same bound; they differ in cost. A nonbasic variable that must move to its opposite bound costs the primal warm start an iteration; relocation moves it without one, but changes the values of the basic variables, which may then violate their bounds and cost the dual method its own iterations. On systems of equations the rows are narrow slabs: measured inside the search (one seed, 60 s, on 165 systems and 140 optimization problems), the slack interval of a row covers on average 13% of the range of the row over the box, against 48% on optimization problems, and a median of 1.9 nonbasic variables per LP change side, against 0.5. SC therefore uses the primal warm start in the optimizer, where the dual one is 4% slower, and the dual one on systems of equations, where both cost about the same; with `compo`, where neither wins on every system, it alternates the two on the first nodes and keeps the one with the lower measured CPU time per LP, trying the other again every 25 nodes (so the choice, unlike the rest of SC, depends on timing).

4.2 Simplex iterations

We follow the bounded-variable simplex of [16]. Each iteration exchanges one basic and one nonbasic variable using the pivot column $\alpha_q = B^{-1}\bar{A}_{:q}$ of the entering variable (FTRAN), the pivot row $\alpha_r^T = \rho_r^T \bar{A}$ with $B^T\rho_r = e_r$ (BTRAN), and updates $x_B \leftarrow x_B - \theta_p\alpha_q$, $\lambda \leftarrow \lambda + \theta_d\rho_r$ and $d \leftarrow d - \theta_d\alpha_r$ (line 27 of Algorithm 2), so that λ, which the certificate needs, is always at hand.

Dual steps. In a dual iteration (lines 10–15 of Algorithm 2) the basis is dual feasible and some basic variable violates its bounds. The leaving variable is the basic variable with the largest violation. The entering variable comes from the ratio test over the pivot row: among the nonbasic z_j whose reduced cost would change sign in the wrong direction, the one with the smallest $|d_j/\alpha_{rj}|$ keeps the basis dual feasible; among the candidates within a relative band of that ratio we take the largest $|\alpha_{rj}|$ (Harris' rule), which avoids tiny pivots. If no entering variable exists, the LP is infeasible and ρ_r is a Farkas ray (line 14).

Rules left out. We also implemented the standard refinements of the dual method [16]: devex pricing [13] for the leaving variable and the long-step (bound flipping) ratio test [18]. On average they did not pay (Section 6), and SC keeps the rules just described, the largest violation and Harris' ratio test: on the small LPs of a pass, the choice of pivots matters less than the work that surrounds them.

Primal steps, tolerances. Primal steps (lines 16–23) enter the nonbasic variable with the largest reduced cost of the wrong sign (Dantzig's rule; with `compo`, whose primal warm starts are longer, devex weights); pivots below a stability threshold are filtered out before choosing the leaving variable, since without this filter a rejected pivot was indistinguishable from optimality. If the entering variable reaches its other bound before any basic variable reaches one, it moves there without a pivot (a bound flip). The feasibility tolerances are 10^{-9}, relative to the value tested, and 10^{-6} with `compo` on systems of equations, whose nearly parallel rows make the LPs degenerate. A tolerance is a matter of efficiency, not of rigor: too wide, it weakens the contraction; too tight, the warm-started simplex reports infeasibility on boxes that are not empty, with rays that do not certify it.

Factorization. SC keeps a sparse LU factorization of B, computed by the factorization code of SoPlex and updated after each pivot by the Forrest–Tomlin method [10], which reuses the FTRAN

of the entering column. It refactorizes every 50 pivots (line 30) and recomputes x_B every 32, to limit drift. With one slack per row, the basis of SC has dimension m, the number of rows: SC works in what SoPlex calls the column representation.

5 Experimental evaluation

A ladder of configurations. All configurations run inside the *same* contractor: the same pass over the $2n$ bounds, with the same skipping rule, the same certificate $\gamma = \bar{A}^T\lambda$ applied to whatever multipliers the engine returns (its dual solution, or its Farkas ray if it reports infeasibility), and the same policy when an LP ends without an optimum: the multipliers where it stopped certify a weaker bound, and the pass continues (Section 3.2). The configurations differ in one step at a time:

R0 *SoPlex as Ibex uses it*: SoPlex 4.0.2 through the LP interface of Ibex, which stores the n variable bounds as rows, with presolving on, the tolerance 10^{-9}, the nearest bound first as in the contractor of Ibex, and at most 100 iterations per LP;

R1 *+ merging twin rows* (Section 3.3);

R2 *+ a tuned SoPlex*: its own instance holding the box as column bounds, presolving off, the tolerance 10^{-6} for systems of equations and 10^{-9} for optimization;[1]

R3 *the bare simplex*: the simplex of SoPlex driven directly through its `SPxSolver` class, the LP kept loaded, only the objective and the column bounds changed between LPs, without the checks, copies and unscaling of the `SoPlex` class; `SPxSolver` itself still refactorizes before declaring an optimum and checks the precision of its solution. It is the cheapest way to use SoPlex. As in R2, the rule of SoPlex chooses the representation of the basis: by columns (dimension m), or by rows (dimension n) when the relaxation has more than 1.2 rows per variable;

R4 *SC* (Section 4);[2]

R5 *the hybrid*: SC on relaxations of at most 30 rows, the bare simplex beyond.

Each rung is compared with R0. HiGHS 1.15.1 and Gurobi 12.0.3, with one thread, the dual simplex, the basis kept between LPs and no presolve, cost 2.3–4.6 times as much per pass as the tuned SoPlex, so we leave them out of the ladder.

Setting. Contractor and solvers are implemented in Ibex 2.8.9, with the affine and `compo` relaxations of Section 3.1. Optimization problems use the affine relaxation, since in preliminary experiments `compo` cost more than it saved with every engine; systems of equations are evaluated with both, since `compo` is the better one on large systems [2] and the affine one keeps most relaxations small. Per node, after merging, the affine relaxation has a median of 5 rows (quartiles 3–9)

[1] Each of these choices is the best one for SoPlex on these LPs: presolving each new relaxation costs more than it saves on LPs of a few pivots, 10^{-9} makes SoPlex report spurious infeasibility on relaxations with nearly equal rows, and 10^{-6} weakens its dual on optimization problems. Turning off scaling, or forcing the row or column representation or the primal algorithm, did not help, and SoPlex 7.1.6 was 1.2–1.9 times slower than 4.0.2 on a sample of eight problems.

[2] On systems with the affine relaxation SC keeps the tolerance 10^{-9} while R2 and R3 use 10^{-6}. With 10^{-6} a pass of SC costs the same (0.84 instead of 0.85 times that of the bare simplex, geometric mean over the 36 systems) and contracts 1% less.

on optimization problems and 16 (9–30) on systems of equations; `compo` has 56 (36–94) on systems. We call an instance, with a given relaxation, *small* if its merged relaxation has at most 30 rows on average over the nodes, and *large* otherwise; these are the two size classes of the tables. For systems of equations we use the solver composition of Ibex (HC4, ACID, interval Newton when the system is square, and the linear contractor within a fixed point with HC4) with the LSmear bisector, on the 242 systems of the Ibex solver benchmark that some configuration solves within 3600 s, 2 seeds, 60 s per run; for optimization, IbexOpt on the 240 problems of the Ibex optimization benchmark that some configuration solves within 3600 s, 4 seeds, 60 s per run. Three of them, bearing, alkyl and alkylbis, are left out of the means: their narrow equalities make the upper bounding of IbexOpt, not the contractor, decide the search, whose tree changes by two orders of magnitude between seeds. All runs were made on an Intel Xeon E5-2620 v4 at 2.1 GHz, with at most 6 GB per run and several runs at a time; times are CPU times.

5.1 Cost of a pass

To measure the contractor without the noise of the search, we built a bank of passes: for 110 instances, 36 systems of equations (one per family) and 74 optimization problems (one per family and size class, among those on which R0 takes more than 1 s), we recorded the input boxes of 1 000 calls of the contractor spread along a search with SC, with the affine relaxation, and replayed every call with each rung from the same input box and relaxation. The affine relaxation exceeds 30 rows on only six of these systems, so a second bank studies large relaxations: it does the same with `compo`, the best relaxation on large systems, for 41 systems whose relaxation has more than 30 rows (31 to 687, median 63; one per family, R0 above 1 s). The 74 optimization problems do not count three that were left out: on pentagon and ex8_2_4, SoPlex as Ibex uses it aborted with an internal exception, and on concon the affine linearization failed for every rung. Two choices of the bank should be kept in mind: its boxes are those visited by a search with SC, although every rung replays them from the same input, and its instances are those on which R0 is slow enough to be measured. Table 1 gives the time per pass of each rung relative to R0. For each instance we normalize both times by the larger one, $t(A)/\max(t(A), t(\mathrm{R0}))$ and $t(\mathrm{R0})/\max(t(A), t(\mathrm{R0}))$, average each over the instances, and report the quotient of the two means; the normalization keeps a few long instances from dominating.

On optimization problems and on systems of up to 30 rows, a pass of SC costs 0.39–0.42 times that of R0, and 25% and 20% less than the bare simplex (SC is cheaper on 70 of the 74 optimization problems and 25 of the 30 systems). The hybrid differs from SC on one optimization problem, of 33 rows. Before SC, merging saves 7–16%, tuning SoPlex 21–27% of what remains, and driving its simplex directly 23–24% more. The median pass on optimization problems takes 264 μs with R0, 136 with the bare simplex and 88 with SC; on small systems, 199, 109 and 85 μs (medians weigh the many fast instances, where SC gains most).

Where the gain comes from. SC does not pivot less: the tuned SoPlex and the bare simplex need a median of 1.8 pivots per LP, SC 2.8, because SoPlex often ends an LP at a vertex that is already optimal for the next bound (Section 6). Each LP of SC is cheaper instead: 11.6 μs on optimization problems and 16.3 on small systems, against 16.2 and 21.6 for the bare simplex. Two experiments locate the difference.

Against the tuned SoPlex (R2), we re-inserted into SC, at every LP, the work that a call of R2 performs, on separate vectors so that the contraction does not change. A pass of SC then goes from 0.55–0.60 to 0.88–0.92 times that of R2 (geometric means over 100 of the 104 instances; the

Table 1: Time per pass of each rung over that of SoPlex as Ibex uses it (R0) on the bank of passes: quotient of the means of the times normalized per instance (see text). Optimization problems and systems of up to 30 rows use the affine relaxation (one optimization problem has 33 rows); systems above 30 rows use `compo`. Last rows: SC over R3 directly, and instances where SC is cheaper than R3.

	optimization	systems of equations	
rung	(74)	≤ 30 rows (30)	> 30 rows (41)
R1 + merging twin rows	0.93	0.84	0.85
R2 + tuned SoPlex	0.68	0.66	0.69
R3 bare simplex	0.52	0.51	**0.55**
R4 SC	**0.39**	**0.42**	0.79
R5 hybrid	0.39	0.42	0.55
R4 / R3	0.75	0.80	1.39
R4 / R3, geometric mean	0.74	0.78	1.42
SC cheaper than R3	70 / 74	25 / 30	4 / 41

emulation failed on four): refactorizing at every LP accounts for 49–56% of the gap (in logarithm), verifying the solution for 22–23%, and recomputing the vectors and copying the solution for 7–8%.

Against the bare simplex (R3), which already skips the checks and copies of the `SoPlex` class, these are not what matters. We added to `SPxSolver` two switches that skip its refactorization before an optimum and its precision check: the factorizations per LP drop from 0.79 to 0.10 on ex7_2_3, yet a pass of R3 becomes only 7% cheaper on optimization problems and 4% on systems, with the same contraction, and SC remains 22% and 12% cheaper (geometric means; it wins on 69 of the 74 problems and 22 of the 36 systems). On bases of 5 to 16 rows a factorization is cheap. What remains is the cost of a call of `SPxSolver::solve` as such: it prepares every LP as a new problem, rebuilding its primal and dual vectors and its pricing and ratio-test vectors from the factorization (4–25% of its time in profiles of four problems), with a per-pivot bookkeeping built for general LPs. Between two LPs, SC only recomputes λ, d and x_B (Algorithm 2).

Merging pays more with SC. Ordered as a ladder, each step is measured on the engine of its rung, and merging is measured on R0, the most expensive one. On the same bank, merging saves 7% of a pass of R0 on optimization problems and 16% on systems of equations, but 21% and 44–47% of a pass of SC. A call of SoPlex carries a fixed cost that does not depend on the rows; SC has almost none, so its time follows the size of the LP, and halving the rows of a square system nearly halves it. Merging is a simple preprocessing, not a contribution of this paper, but it is worth most with an engine that has removed the fixed cost of a call.

Same contraction. All rungs empty the same boxes and contract alike: the sum of the widths of the box shrinks by 25.5% per call on average on optimization problems and by 18.6–18.7% on small systems for R1 to R4, and within 1% on all but 6–9 of the 110 instances. The exception is R0, which ends 8.5% of the LPs of small systems at its limit of 100 iterations (0.9% after merging; the tuned SoPlex, with the same limit, almost none); the multipliers where it stops certify a weaker bound, and R0 contracts by 18.2% (on Geneig-icse, the extreme case, by 23% against 34%). On the `compo` bank the LPs are larger and more degenerate, and with the tolerance of 10^{-6} the certified

bounds of the engines differ slightly more: SC contracts by 15.6%, the tuned SoPlex by 15.9% and the bare simplex by 16.1%, while the limit of R0 ends 5% of the LPs without an optimum.

Large relaxations. On the systems whose relaxation exceeds 30 rows the order of the last two rungs is reversed (Figure 1): on the `compo` bank a pass of SC costs 1.39 times that of the bare simplex (1.42 in geometric mean) and is cheaper on only 4 of the 41 systems, and on the six affine ones it costs 1.49 times (geometric mean) and loses on all. Merging and tuning SoPlex keep their weight; the hybrid hands these relaxations to the bare simplex.

Why SC is slower there. What SC saves is a fixed cost per LP, and on these relaxations an LP needs a dozen pivots instead of two or three, so the saving no longer weighs. What remains is the cost of the pivots, and three different things happen there.

The representation of the basis is the main one. A vertex of the relaxation can be described by a basis of dimension m, with one basic column per row, or by a basis of dimension n, the n rows and bounds that are tight. SC, with one slack per row, always works with the first. SoPlex chooses: by columns when the relaxation has at most 1.2 rows per variable, by rows otherwise. With `compo` the rows outnumber the variables several times (147 rows for 35 variables on Fourbar-icse), so the bare simplex factorizes and solves with a basis of dimension n where SC uses one of dimension m: with almost the same pivots (1.05 times), SC spends there three times as long in the triangular solves of the entering column. To measure how much of the gap this explains, we forced the bare simplex to the column representation on the `compo` bank. It becomes 1.35 times slower (geometric mean), with the same contraction, and a pass of SC then costs 1.04 times that of the bare simplex instead of 1.41 in the same runs, and is cheaper on 18 of the 41 systems; on the systems without the two difficulties below (the Redeco and Eco families, Katsura-30, Dietmaier, Directkin, Butcher) the two are within 10% of each other. The representation thus accounts for almost 90% of the gap, in logarithm. On the bank of the affine relaxation, where merging leaves about as many rows as variables, SoPlex already works by columns, and forcing it to rows makes it slower (a pass of SC goes from 0.85 to 0.79 times that of the bare simplex).

Stalling remains on a few systems: on Katsurabis-14, 17 of the 16 562 LPs of SC, degenerate ones, pivot until the limit of 100 iterations, while the bare simplex reaches an optimum on all of its LPs. The primal steps of SC, with Dantzig's rule, have no device against stalling, such as the bound perturbation that SoPlex applies on degenerate steps.

The systems with band structure (Bratu, BroydenTri, BroydenBanded, DiscreteBoundary) are the third case: there SC makes several times the pivots of the bare simplex, up to eleven times on the six affine systems above 30 rows, where both engines work by columns. Devex pricing and the long-step ratio test (Section 4.2) help on them with the affine relaxation: they bring a pass of SC from 0.71 to 0.55 times that of R0 on those six systems (the bare simplex: 0.47), but leave it at 1.51 times the bare simplex on the `compo` bank (geometric mean), and change little on the smaller relaxations. We have no explanation for these systems beyond the pivoting rules of SoPlex, its automatic pricer with the bound-flipping ratio test.

5.2 The whole solver

Table 2 runs the same ladder inside the search. To weight families equally, we keep one instance per family and size class, the one with the largest relaxation; families are the prefix of the name on systems of equations, while on optimization problems every problem is its own family, except for reformulations of the same problem (`bis`, `-1`). To populate both size classes of the systems, we

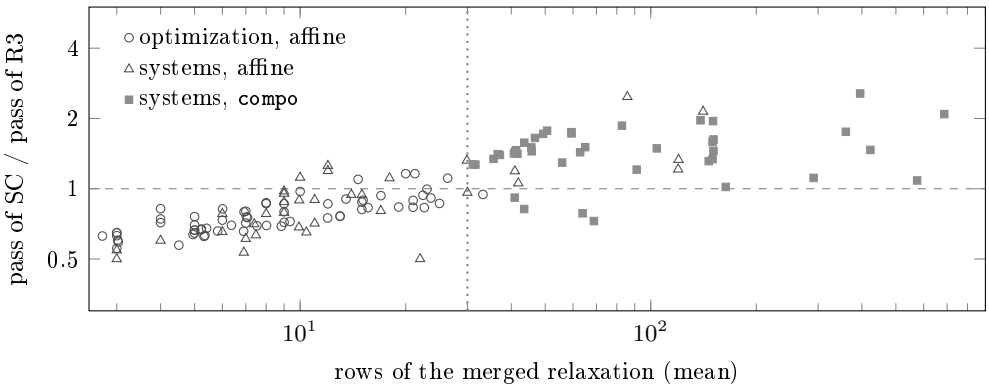


Figure 1: Time per pass of SC over that of the bare simplex (R3) on the two banks, one point per instance, against the size of the relaxation. Below the dashed line SC is cheaper; the dotted line is the threshold of the hybrid.

Table 2: CPU time of each rung over that of SoPlex as Ibex uses it (R0), inside the search, one instance per family and size class, over the runs where one of the two configurations takes more than 1 s (resp. 5 s); in parentheses, the families of each class before that filter. On the classes up to 30 rows the hybrid is not reported: it runs SC on almost every node. Systems of equations pool the affine and `compo` relaxations. The search trees of R1 to R5 agree within 3%.

	optimization		systems of equations			
	≤ 30 rows (189)		≤ 30 rows (65)		> 30 rows (52)	
rung	> 1 s	> 5 s	> 1 s	> 5 s	> 1 s	> 5 s
R1 + merging twin rows	0.97	0.95	0.96	0.95	0.91	0.89
R2 + tuned SoPlex	0.90	0.90	0.89	0.87	0.85	0.82
R3 bare simplex	0.84	0.85	0.85	0.84	0.79	0.76
R4 SC	**0.78**	**0.79**	**0.83**	**0.83**	0.91	0.87
R5 hybrid	–	–	–	–	**0.77**	**0.75**

pool their runs with the affine and `compo` relaxations, each (instance, relaxation) classified by its own size. Times are normalized per run as on the bank; unfinished runs count as 60 s.

The ladder keeps its shape, but the gains are diluted: the pass is only part of the cost of a node, with the linearization, HC4, ACID and interval Newton. On optimization problems the solver takes 21–22% less CPU time with SC than with R0, and 5–8% less than with the bare simplex. On small systems SC and the bare simplex nearly tie (0.83 against 0.84–0.85), because the pass weighs less in the system solver. Counting the calls of the contractor in a search with SC and multiplying them by the cost of a pass on the bank, the pass takes a median of 16% of the CPU on optimization problems (interquartile range 7–26%) but 5% on systems of equations (2–13%), where the linearization, HC4, ACID and interval Newton dominate a node; saving 20% of 5% is about 1%. On large systems SC is 15–16% slower than the bare simplex, and the hybrid, which runs the bare simplex there, matches it (0.98). Only three optimization families exceed 30 rows (33–49 rows); SC remains the best on them (0.82 against 0.97 for the bare simplex), so the hybrid could keep SC on optimization problems at any size. Devex pricing and the long-step ratio test change these ratios by at most 0.01 up to 30 rows, and make SC 4–6% slower on large systems.

The threshold of the hybrid. From the runs of every rung we evaluated offline the hybrid that takes, for each (instance, relaxation), SC if its mean relaxation has at most T rows and the bare simplex otherwise. Against the bare simplex, SC gains below 15 rows (0.86–0.92 on optimization problems, 0.95–0.98 on systems), ties from 15 to 30, and loses from 40 rows on (1.05 at 40–50 rows, 1.15 beyond 120 on systems). The total time of the hybrid is flat for T between 15 and 30, and we use $T = 30$; the contractor applies it per node, with the size of the relaxation of the previous node. Run inside the search on the systems of equations with `compo` (two seeds), the hybrid with $T = 15$, 20 and 30 costs 0.98–1.00 times the bare simplex on the small systems and 0.98 on the large ones, where SC alone costs 1.15: the choice per node loses nothing, and the threshold does not matter inside this range.

6 Discussion

Why SC wins. Every rung solves the same relaxation and feeds the same certificate, and from R1 on they contract alike: the gain is the cost of an LP. The ladder separates what a careful integration of SoPlex removes, the interface and its options (R1 to R3), from what only a simplex dedicated to the pass can omit, the preparation of each LP as a new problem. The certificate makes the omission possible: any basis certifies and can be made dual feasible, so between two LPs only what the new objective changes has to be recomputed.

Where it loses. Beyond 30 rows the pivots dominate the cost of an LP, and SoPlex makes them cheaper by representing the basis by rows when the rows outnumber the variables (Section 5.1). The homogeneous form, with one slack per row, which makes the certificate and the warm starts simple, is what ties SC to the column representation; a self-certifying simplex in the row representation would keep the certificate, which needs only the multipliers, and is the natural next step. Meanwhile the hybrid uses each engine where it is better. On small relaxations one difference remains unexplained: SoPlex ends 13–40% of its LPs on optimization problems at a vertex that is already optimal for the next bound, and needs fewer pivots than SC (1.8 against 2.8 per LP); neither a tie-break toward the corners of the box inside the optimal face nor a short walk inside that face made SC end at such vertices.

What did not pay. The refinements of the dual method left out of SC (Section 4.2), devex pricing and the long-step ratio test, help on some relaxations and hurt on others: on the bank they make a pass 1.3–1.5 times faster on three of the band systems with the affine relaxation, and up to 2.1 times slower on the same systems with `compo`, and on average they leave the time of the search unchanged (Section 5.2). Several ideas that use the certificate or the tableau beyond the pass did not pay on the whole benchmark either. Certificates of an ancestor box remain valid (Proposition 3) but rarely prune what the node does not prune anyway. Truncating each LP when the certified bound of its current basis (Proposition 5) is close to what the LP can still gain saves pivots ($\times 0.84$) but weakens the contraction (nodes $\times 1.39$–1.60). Propagating with HC4 in the middle of a pass, cuts at the LP optima, and a Lagrangian bound improved by monotonicity reduce the nodes or tighten bounds, but cost more than they save. Inheriting the basis of the parent node cuts the pivots of the first LP of a node by a factor of three or more, but the first LP is a small part of the node. Earlier versions of SC kept the whole tableau explicitly and perturbed the zero reduced costs of the highly dual-degenerate bound problems [16]; with the other components in place both made SC slower. Doing more work inside a node did not change the search tree; what changed it

was what is computed: the stability filter of the ratio test, the Farkas ray of the infeasible row, the tolerance.

Limits. SC does not remove SoPlex entirely: it factorizes its basis, and updates the factors, with the LU code of SoPlex (Section 4.2). Relaxations of several hundred rows are still expensive for every engine: the systems of 400 to 1 000 variables of the benchmark exceed the limits of 60 s and 6 GB with all of them. And the gain of SC on the whole system solver is small, because the pass is a small part of its nodes (5% of the CPU in the median).

7 Conclusion

A polytope-hull contractor solves $2n$ small, closely related LPs per node and certifies each bound with the Neumaier–Shcherbina bound, which is valid for any multipliers. A general-purpose LP solver cannot take advantage of this: it treats each LP as a new problem. A simplex dedicated to the pass solves the $2n$ problems in sequence on one basis, updating only what changes from one LP to the next; the certificate makes it simple, since it never verifies or refactorizes before an optimum and needs no phase 1. Switching off the refactorization and the checks in SoPlex recovers little of the difference: the gain is in not preparing each LP as a new problem. Measured as a ladder inside the same contractor, from the same boxes and relaxations, a pass of this simplex costs 0.39–0.42 times that of SoPlex as Ibex uses it on relaxations of up to 30 rows, and 20–25% less than the simplex of SoPlex driven directly, with the same contraction. Inside the search, the optimizer takes 21–22% less CPU time than with SoPlex as Ibex uses it, and 5–8% less than with the bare simplex. On larger relaxations the pivots dominate, and when the rows outnumber the variables SoPlex makes them cheaper by representing the basis by rows: against SoPlex in the column representation, the one of SC, a pass of SC is within 4%. A hybrid that chooses the engine by the size of the relaxation keeps the best of both. A self-certifying simplex in the row representation is the natural next step.

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Code and data availability

The code and data used in the experiments are archived at Zenodo (**[TODO:** *DOI*]); they are the branch `dfb-sc-paper` of `https://github.com/rilianx/ibex-lib`. SC and the contractor are in `examples/dfb`, and a standalone version of SC that builds against an installed Ibex is in `examples/sc_plugin`; `results/escalera_2026-11-06` holds the banks of passes (input boxes and relaxations), the raw results of every run, the scripts that ran them and those that produce the tables, with instructions to rebuild the binaries. The fix to the polytope-hull contractor of Ibex used by every configuration was submitted upstream (`https://github.com/ibex-team/ibex-lib/pull/579`).

Use of generative AI

The authors used Claude (Anthropic), through the Claude Code assistant, to help implement the contractor and the simplex, to write the scripts that run and analyze the experiments, and to draft

and edit the text. The authors directed the work, checked the code, the results and the text, and take full responsibility for the content of the paper.

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